i) When [S] in very much less than K_m:

The Michaelis-Menten equation is

$$v = \frac{(V_{max}[S])}{(K_m + [S])}$$

When [S] << K_m; the equation becomes

$$v = \frac{(V_{max}[S])}{(K_m)}$$

When both the constants 'V_{max}' and 'K_m' are replaced by another constant 'K' the above equation becomes v = K[S]

i.e. the velocity is directly proportional to the substrate concentration (first order) (Fig 9.4)

ii) When [S] is very much greater than K_m:

$$v = \frac{(V_{max}[S])}{(K_m + [S])}$$

When [S] >> K_m the equation becomes,

$$v = \frac{(V_{max}[S])}{([S])} = V_{max}$$

The velocity is independent of substrate concentration (zero order) (Fig 9.4)

iii) When [S] = K_m : (Derivation of K_m):

$$v = \frac{(V_{max}[S])}{(K_m + [S])}$$

When K_m = [S]

$$v = \frac{(V_{max}[S])}{([S]+[S])} = \frac{(V_{max}[S])}{(2[S])} = ½\ V_{max}$$

Thus K_m = [S] at ½ V_{max}

9.1.2. Significance of K_m : (Michaelis-Menten Constant)

K_m is defined as the substrate concentration at half maximal velocity.

At ½ V_{max}, K_m = [S]

K_m is also defined as

$$K_m = \frac{(k_2 + k_3)}{k_1}$$

k_1 is the rate constant for the formation of ES complex,

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